



UNIVERSITY OF RWANDA

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A Fractional-Order Compartmental Model of HIV Transmission with Social Behaviour Interventions

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Final Year Project Report

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Background / Motivation

Rwanda-aware public-health motivation

- ▶ HIV transmission is shaped by biology, treatment access, and social behaviour.
- ▶ Awareness, safer behaviour, testing, treatment-seeking, and adherence can change long-term outcomes.
- ▶ In Rwanda, continued prevention and adherence support remain important alongside treatment progress.

Why mathematics? A model gives a controlled way to compare intervention strategies and understand possible long-term behaviour.

Context

HIV in Rwanda

Mechanism

Social behaviour

Method

Fractional memory

Output

Simulation evidence

Problem Statement

Limitations in ordinary modelling

- ▶ Ordinary models usually depend only on the present state.
- ▶ HIV-related behaviour may depend on past awareness, stigma, testing history, and adherence.
- ▶ Social behaviour is often simplified into fixed rates.

Need addressed by this project

- ▶ Include memory using a Caputo fractional derivative.
- ▶ Represent behavioural interventions through effective rates.
- ▶ Use numerical simulation to compare scenarios clearly.

Research focus: How do fractional memory and social behaviour interventions affect simulated HIV dynamics in a SITA framework?

Literature Review and Research Gap

What existing studies show

- ▶ Classical HIV models commonly use ordinary differential equations and treatment compartments such as SICA/SITA.
- ▶ Previous studies show that social behaviour, testing, safer sexual practices, stigma, and adherence influence HIV transmission.
- ▶ Fractional-order epidemic models use Caputo derivatives to capture memory and hereditary effects.

Research gap

Existing work does not commonly combine:

- ▶ fractional HIV dynamics,
- ▶ social behaviour intervention functions,
- ▶ and an interactive simulation dashboard.

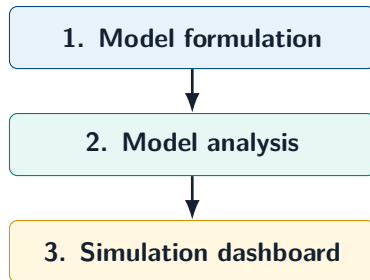
Gap addressed: a fractional-order SITA model that connects memory effects, social behaviour interventions, numerical simulation, and dashboard exploration.

Objectives

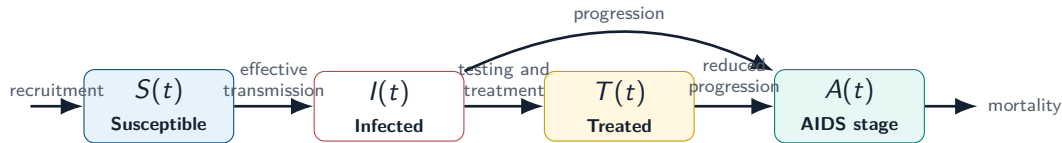
Main objective

To formulate, analyse, and computationally simulate a fractional-order SITA model of HIV transmission incorporating social behaviour interventions. **Specific objectives**

1. Formulate a fractional-order SITA HIV model using the Caputo derivative and social behaviour interventions.
2. Analyse positivity, boundedness, disease-free equilibrium, $\mathcal{R}_0$, and fractional stability.
3. Implement simulations and a Python dashboard to study memory effects, interventions, sensitivity, and epidemic trajectories.



Model Diagram: SITA



State variables: $S(t)$ susceptible, $I(t)$ infected, $T(t)$ treated, $A(t)$ AIDS-stage population.

Controls: u_1 awareness, u_2 safer behaviour, u_3 testing, u_4 adherence.

Fractional-Order Model

Caputo memory form

$${}^C D_t^q X(t) = F(t, X(t)), \quad 0 < q \leq 1$$

$${}^C D_t^q S = \Lambda - \beta_{\text{eff}} S(I + \eta T) - \mu S,$$

$${}^C D_t^q I = \beta_{\text{eff}} S(I + \eta T) - (\tau_{\text{eff}} + \delta + \mu) I,$$

$${}^C D_t^q T = \tau_{\text{eff}} I - (\rho_{\text{eff}} + \mu) T,$$

$${}^C D_t^q A = \delta I + \rho_{\text{eff}} T - (d + \mu) A.$$

Intervention-adjusted rates

$$\beta_{\text{eff}} = \beta_0(1 - u_1)(1 - u_2)$$

$$\tau_{\text{eff}} = \tau(1 + u_3)$$

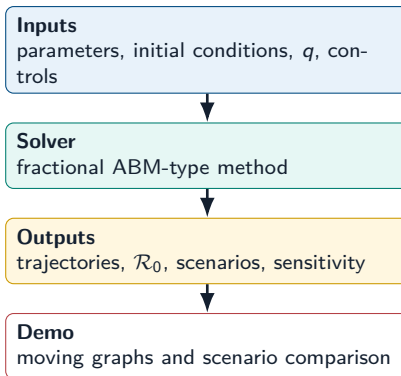
$$\rho_{\text{eff}} = \rho(1 - u_4)$$

When $q = 1$, the model is ordinary. When $q < 1$, past states influence current change.

Numerical Scheme + Implementation

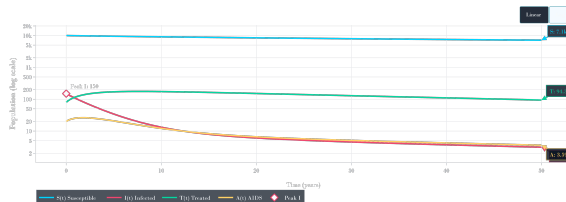
Numerical idea

- ▶ Time interval is divided into discrete steps.
- ▶ A predictor estimates the next state.
- ▶ A corrector updates the state using fractional-memory weights.
- ▶ One engine supports baseline, memory, intervention, and sensitivity experiments.



Presentation link: the slides explain the model; moving graphs are shown separately during the live demonstration.

Baseline Simulation Result

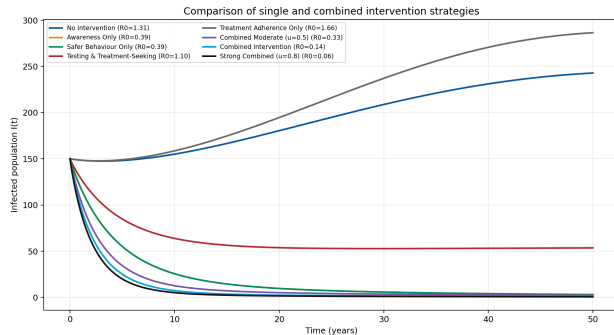


Computed baseline

- $\mathcal{R}_0 = 0.430 < 1$, below the epidemic threshold.
- Peak infected population is 150 at $t = 0$ years.
- Final values after 50 years: $I = 3.1$, $T = 94.3$, $A = 3.6$.

Interpretation: the baseline intervention-adjusted configuration moves toward disease-free behaviour.

Simulation Results: Intervention Comparison

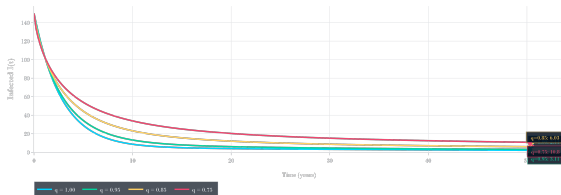


Main interpretation

- ▶ Single interventions help, but each targets only one mechanism.
- ▶ Combined intervention acts on transmission, treatment uptake, and progression.
- ▶ The combined strategy gives stronger simulated control than isolated changes.

Public-health meaning: prevention and treatment support should work together.

Simulation Results: Fractional Memory

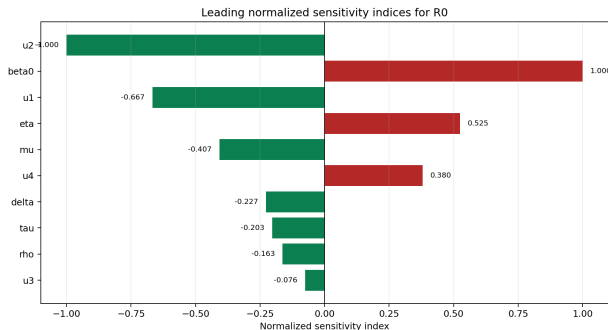


Why q matters

- ▶ $q = 1$ represents the ordinary model.
- ▶ Smaller q values strengthen the memory effect.
- ▶ Different q values produce visibly different trajectories.

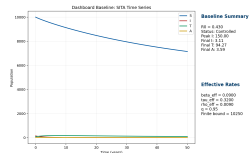
Defense point: fractional calculus is not decorative; it changes the simulated dynamics.

Dashboard and Sensitivity Evidence



Main evidence

- ▶ Transmission-related quantities dominate R_0 sensitivity.
- ▶ Safer behaviour and awareness strongly reduce R_0 .
- ▶ Dashboard outputs support thesis discussion and live defense.



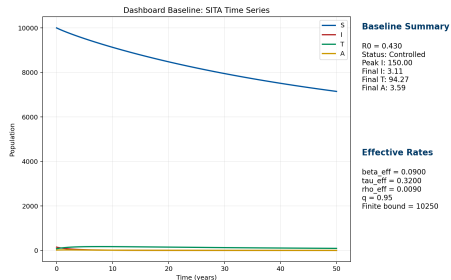
Live Dashboard Demonstration

Open during the defense

Dashboard link

<https://hiv-model.onrender.com/>

1. Run the baseline simulation.
2. Show the final state summary and interpretation.
3. Compare no intervention with combined intervention.
4. Change q to show the fractional memory effect.



After the demo: return to the next slide for conclusion and recommendations.

Conclusion + Recommendations

Conclusion

- ▶ The SITA HIV model was extended using fractional memory.
- ▶ Behavioural interventions were represented through effective rates.
- ▶ Numerical simulations support the value of combined intervention strategies.

Recommendations

- ▶ Calibrate parameters with Rwanda-specific data.
- ▶ Extend the model using age, gender, or stochastic structure.
- ▶ Compare other fractional operators.
- ▶ Improve the dashboard into an educational decision-support tool.

Final message: the work connects mathematical analysis, numerical simulation, and a working dashboard for communicating HIV intervention dynamics.